Short - Answer Test

Name:- K. Rama keerthi

Reg No:- 192312495

Course Code:- ITA0502

Course Name:- Computer Vision.

1-A Sampling:-

sampling is the process of converting a continuous image into a discreate set of pixels.

To improve the image, we can increase the sampling rate / spatial resolution.

* Increasing a smartphone camera from 2MP to 18MP gives more pixels.

2-A Dark and Noisy Image

For a dark and noisy night image, image enhancement and noise filtering can be applied.

Example:- enhancing a CCTV night image makes a person's face more visible.

3-A Quantization

Quantization converts continous intensity values into a limited no. of discreate digital values.

It is an important step in converting a real-world image into digital image.

Example:-
A medical X-ray can be converted into digital intensity levels for computer analysis.

4-A Image enhancement:-

Image enhancement improves the visual quality of an image and makes important details clearer.

Example:-
In a factory, scratches on a mental component become easier to identify after enhancement.

5-A Image segmentation:-

Image segmentation divides an image into meaningful regions or objects. This helps the vehicle understand its surroundings and make safe decisions

6-A Edge detection

* edge detection identifies sudden change in intensity and detects boundaries objects.
* To detected edges help locate the shape and position of defects.

example:- A crack on a manufactured mental component appears an edge.

7-A Image filtering

Image filtering is used to remove unwanted noise from an image.

* It makes the medical image smoother and cleaner.

Example:-

filtering an ultrasound or medical image

8-A Object detection:-

* object detection identifies objects in an image and determine their locations.
* In traffic monitoring, it can detect and

track vehicles.

eg:- Traffic camera detects the number and movement of cars on a busy road

9-A Motion Blur:-

The blur is caused by movement of the object or camera during image acquisition.

* This is called motion blur and causes the image details to appear unclear.

Example:-

A moving a car.

10-A Image classification

Image classification assigns an image to a particular category.

* It In an industrial system, products can be classified as defective or good.

Example:-

A camera-based system can classify a bottle with a crack.